

Linear Regression

Alexander McLain

June 2, 2026

Outline

Introduction

Simple Linear Regression (SLR)

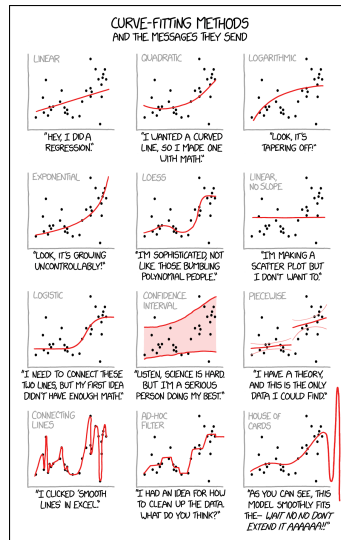
Estimating SLR

Inferences

Multiple Linear Regression (MLR)

Model Diagnostics

Inference on the MLR Model



Linear Regression

- ▶ Linear regression models predate computers and are still among the most widely used tools in applied statistics.
- ▶ They are simple, interpretable, and often adequate for real-world relationships, especially when sample sizes are small or signals are modest.
- ▶ Two uses, often blurred together:
 - ▶ **Explanation:** estimate the relationship between a predictor and an outcome, adjusting for other factors.
 - ▶ **Prediction:** produce a value of the outcome for new observations.
- ▶ Linear models have assumptions, but they are reasonably robust to mild violations. When assumptions clearly fail, we can transform variables, add flexibility (splines, interactions), or move to a richer model.

SLR Background

- ▶ One numerical outcome, one (or more) predictors.
- ▶ Two roles:
 - ▶ **Outcome** Y (also: response, dependent variable).
 - ▶ **Predictors** $X_1, \dots, X_p$ (also: covariates, independent variables, features).
- ▶ The word “linear” refers to the *coefficients*, not the predictors. Polynomial and interaction terms still count as linear regression because they are linear in β .
- ▶ **Simple linear regression (SLR)**: one predictor.

The SLR Model

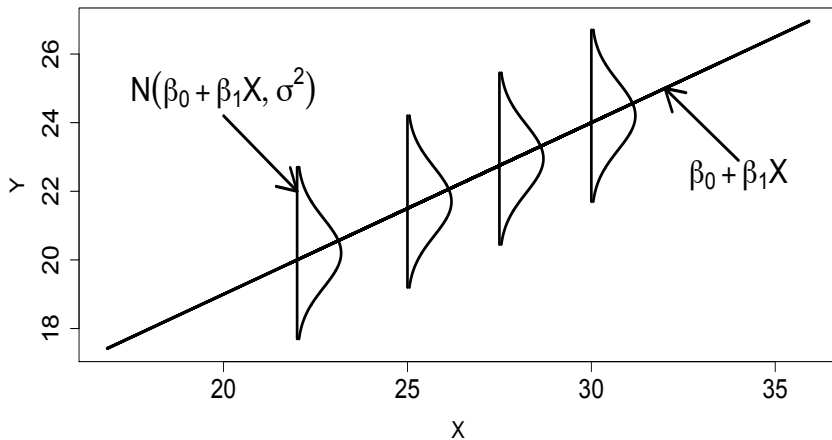
- ▶ Model: $Y = \beta_0 + \beta_1 X + \epsilon$
- ▶ β_0 : intercept. The mean of Y when $X = 0$ (often not meaningful in practice when $X = 0$ is outside the data range).
- ▶ β_1 : slope. The expected change in Y associated with a one unit increase in X .
- ▶ ϵ : random error. The piece of Y that X does not explain. Assumed independent across observations with constant variance σ^2 .
- ▶ In words: at any value of X , the outcome Y scatters around the line $\beta_0 + \beta_1 X$ with spread σ .

Assumptions of Linear Regression: LINE

- ▶ Four assumptions, packaged as the **LINE** acronym:
 - ▶ **L**inearity: the mean of Y is a linear function of X .
 - ▶ **I**ndependence: observations do not influence each other.
 - ▶ **N**ormality: residuals are normally distributed.
 - ▶ **E**qual variance: residual spread does not change with X (homoscedasticity).
- ▶ Linearity and independence are the two that matter most. Mild violations of normality and equal variance rarely change conclusions in moderate-to-large samples.
- ▶ All four are checked using *residual diagnostic plots* after fitting (next major section).

*Less obvious: the X values are treated as fixed. When predictors are themselves measured with error, additional considerations apply.

Regression Assumptions: Picture

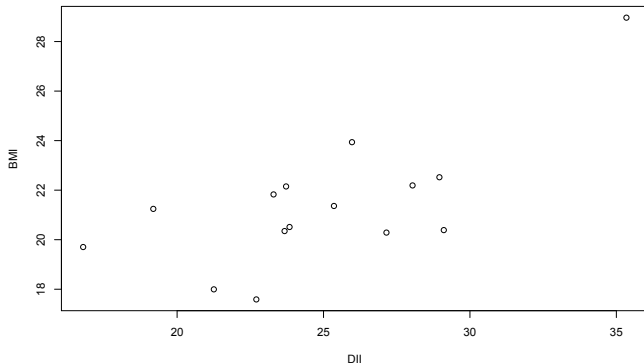


Estimating the Line

- ▶ We do not know β_0 and β_1 . We *estimate* them from data.
- ▶ Estimates are written $\hat{\beta}_0$ and $\hat{\beta}_1$ (equivalently b_0 , b_1).
- ▶ The fitted line is $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$.
- ▶ $\hat{Y}$ serves two purposes:
 - ▶ Best guess for the *mean* of Y at a given X .
 - ▶ Best guess for a *new* Y at a given X (uncertainty is larger here, more on this later).
- ▶ Open question: of all possible lines, which one fits best?

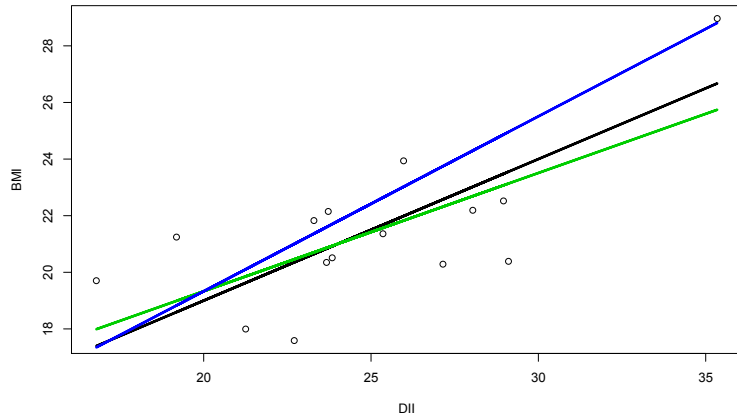
Estimating SLR: Example Data

- ▶ Let $X = \text{Dietary Inflammation Index (DII)}$, $Y = \text{Body Mass Index}$.
- ▶ We have $n = 15$ data points.

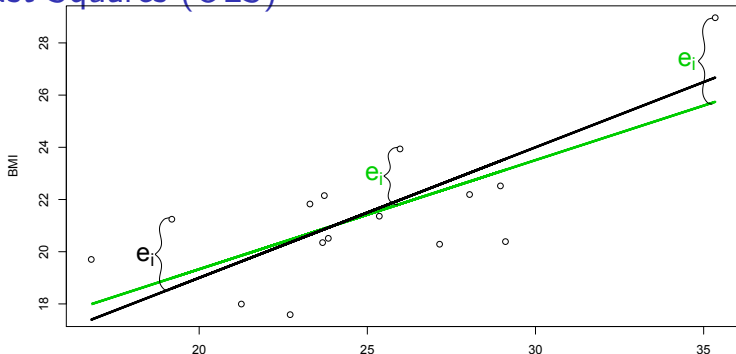


Choosing the Best Line

- ▶ Three candidate lines below. Which is best? Why?



Ordinary Least Squares (OLS)



- ▶ For each candidate line, compute the vertical distance from each point to the line. Square those distances, add them up.
- ▶ **OLS:** pick the line that makes this sum as small as possible.

$$\text{minimize } \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Worked Example: Age and SBP

- ▶ X : Age (years); Y : Systolic blood pressure (mmHg)
Age (x): 38, 41, 45, 48, 51, 53, 57, 61, 65
SBP (y): 116, 120, 123, 131, 142, 145, 148, 150, 152
- ▶ Sample size $n = 9$. Mean age 51 years; mean SBP 136.3 mmHg.
- ▶ Running `lm(SBP ~ Age)` returns:
 - ▶ $\hat{\beta}_0 = 60.5$ (intercept)
 - ▶ $\hat{\beta}_1 = 1.49$ (slope, mmHg per year)
- ▶ Fitted line: $\hat{Y} = 60.5 + 1.49 \cdot \text{Age}$
- ▶ **Interpretation:** each additional year of age is associated with a 1.49 mmHg higher *average* SBP. A 10-year age difference corresponds to about a 15 mmHg difference in average SBP.
- ▶ Predicted SBP at age 55: $60.5 + 1.49(55) = 142.5$ mmHg.
- ▶ **Question:** should we use this line to predict SBP at age 16? Age 88? Why or why not?

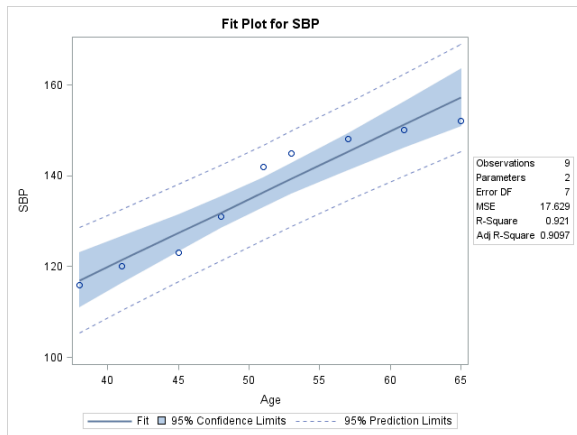
Coefficient of Determination R^2

- ▶ Total variation in Y splits into two pieces: variation *explained* by the fitted line, plus variation left *unexplained* (the residuals).

$$R^2 = \frac{\text{variation explained}}{\text{total variation}}$$

- ▶ Range: $0 \leq R^2 \leq 1$. Higher means the line accounts for more of the variation in Y .
- ▶ For SLR, R^2 equals the squared Pearson correlation between X and Y .
- ▶ **Age and SBP example:** $R^2 = 0.92$. Age alone accounts for 92% of the variation in SBP in this small sample. (This is unusually high because the sample is small and selected to span a wide age range.)
- ▶ **Caution:** R^2 never decreases when you add predictors, even useless ones. Use *adjusted* R^2 , AIC, or formal F-tests for model comparison, not raw R^2 .

Fit of Age–SBP Example



Why Inference?

- ▶ $\hat{\beta}_1 = 1.49$ is a point estimate from *one* sample. A different sample of 9 people would give a different number.
- ▶ Three questions inference can answer:
 1. How much would $\hat{\beta}_1$ vary across samples? Captured by its **standard error**.
 2. What is a plausible range for the true population β_1 ? Answered by a **confidence interval**.
 3. Is β_1 different from zero? Answered by a **hypothesis test**.
- ▶ All three quantities are reported by `summary(lm_object)` in R . We focus on *interpretation*, not the arithmetic.

Confidence Interval for β_1

- ▶ A $(1 - \alpha)100\%$ CI takes the form:

$$\hat{\beta}_1 \pm t^* \cdot \text{SE}(\hat{\beta}_1)$$

where t^* is a critical value from the t -distribution.

- ▶ **Age and SBP:** $\hat{\beta}_1 = 1.49$, $\text{SE} = 0.164$, $95\% \text{ CI} = (1.10, 1.88)$.
- ▶ **Interpretation:** we are 95% confident that in the broader population this sample came from, the true mean increase in SBP per additional year of age is between 1.10 and 1.88 mmHg.
- ▶ The CI does *not* say there is a 95% chance the true slope is in this interval. (The true slope is fixed; the interval is random.) The 95% refers to the long-run coverage of the procedure across hypothetical repeated samples.

Hypothesis Test for β_1

- ▶ $H_0 : \beta_1 = 0$ vs. $H_1 : \beta_1 \neq 0$.
- ▶ In plain English: “is there any linear association between X and Y in the population?”
- ▶ Test statistic: $t = \hat{\beta}_1 / \text{SE}(\hat{\beta}_1)$
- ▶ **Age and SBP:** $t = 1.49 / 0.164 = 9.09$, p-value < 0.0001 . Reject H_0 .
- ▶ **Conclusion:** there is strong evidence that average SBP increases with age in this population.
- ▶ A non-rejection does *not* prove $\beta_1 = 0$. It means the data are consistent with $\beta_1 = 0$ given the sample size. Small samples have low power.

Confidence Interval vs. Prediction Interval

Two different questions, two different intervals at the same X .

- ▶ **Confidence interval (CI):** where is the *average* Y at this X ?
- ▶ **Prediction interval (PI):** where might a *new individual's* Y at this X fall?
- ▶ The PI is always wider, because it adds the spread of individuals around the line on top of the uncertainty in the line itself.
- ▶ **Age and SBP at age 50:**
 - ▶ Point estimate $\hat{Y} = 135.0$ mmHg.
 - ▶ 95% CI for the mean SBP of 50-year-olds: (131.6, 138.3).
 - ▶ 95% PI for one new 50-year-old's SBP: (125.5, 145.4).
- ▶ Use the CI when you want the population average. Use the PI when you are predicting an individual.
- ▶ Both intervals are narrowest at $X = \bar{X}$ and fan out at the extremes. This is why extrapolation is risky.

ANOVA Table for SLR

The ANOVA table organizes total variation into “explained” and “unexplained.”

Source	DF	SS	MS	F
Model	1	SS_{Reg}	MS_{Reg}	F_0
Error	$n - 2$	SS_{Err}	MS_{Err}	
Total	$n - 1$	SS_{Tot}		

Age and SBP example:

Source	DF	SS	MS	F
Model	1	1438.6	1438.6	81.6
Error	7	123.4	17.6	
Total	8	1562.0		

- $F = 81.6$ with p-value < 0.0001 . Strong evidence that the model explains more variation than chance alone would produce.

F-test, t-test, and Correlation

- ▶ For SLR with one predictor, three tests are mathematically equivalent and give identical p-values:
 1. ANOVA F-test for the model.
 2. t-test for $H_0 : \beta_1 = 0$.
 3. Test for $H_0 : \rho = 0$ (Pearson correlation).
- ▶ In MLR (next section) these come apart: the F-test tests *all* slopes jointly, while the t-test for each β_j tests that one slope given the others are in the model.
- ▶ R reports both at the bottom of `summary(lm_object)`: the overall F-statistic and the per-coefficient t-statistics.

Review: What Can We Do with SLR?

- ▶ Fit a regression line to describe the average relationship between X and Y .
- ▶ Report R^2 as a summary of how tightly Y is explained by X .
- ▶ Test whether X is linearly associated with Y .
- ▶ Build confidence intervals for the population mean response at a given X , and prediction intervals for a new observation at that X .
- ▶ **A confounding question.** Suppose X = stress level and Y = severity of muscle cramps. We reject $H_0 : \beta_1 = 0$ with $\hat{\beta}_1 > 0$. What else could explain this association besides “stress causes cramps”?
 - ▶ Reverse causation: cramps cause stress.
 - ▶ A confounder: poor sleep raises both stress and cramps.
 - ▶ Selection: only people who already have both end up in the study.

SLR alone cannot tell you which is going on. MLR helps with *measured* confounders. Reverse causation and selection require study design, not modeling.

Multiple Linear Regression

- ▶ One outcome Y ; p predictors $X_1, \dots, X_p$.
- ▶ Model: $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon$
- ▶ **Interpretation of β_j :** the expected change in Y for a one unit increase in X_j , *holding the other predictors constant*. The italicized phrase is the entire point of MLR.
- ▶ Two distinct goals, sometimes leading to different models:
 1. **Inference:** get an unbiased estimate of a specific β_j of interest, after adjusting for confounders.
 2. **Prediction:** get the smallest expected prediction error on new data, regardless of which variables are in the model.
- ▶ **Trick question:** is $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \epsilon$ SLR or MLR? Why?

MLR Example: VO_2Max

- ▶ **Outcome:** VO_2Max (maximal oxygen uptake, mL/kg/min). A standard measure of aerobic fitness.
- ▶ **Predictors:**
 - ▶ Runtime for 1.5 miles (minutes)
 - ▶ Age (years)
 - ▶ Weight (kg)
 - ▶ Resting heart rate (bpm)
 - ▶ Running heart rate (bpm)
 - ▶ Maximum heart rate (bpm)

MLR Example: VO_2Max

- ▶ **Sanity check before fitting:** what direction should each $\hat{\beta}$ go?
 - ▶ Faster runtime should mean higher VO_2Max (negative slope).
 - ▶ Older age usually means lower VO_2Max (negative slope).
 - ▶ Heavier weight tends to mean lower mL/kg/min (negative slope).
 - ▶ Lower resting heart rate is a sign of fitness (negative slope).
- ▶ If a fitted coefficient has an unexpected sign, that is a red flag worth investigating. It often points to collinearity, a poorly specified model, or a measurement issue.

MLR Assumptions

- ▶ The same LINE assumptions apply: Linearity, Independence, Normality, Equal variance.
- ▶ One additional concern in MLR:
 - ▶ **No perfect collinearity.** No predictor can be an exact linear combination of the others.
 - ▶ Perfect collinearity makes the OLS solution undefined, and R will drop one of the offending variables and warn you.
 - ▶ Common causes: including a variable and its duplicate; including all levels of a categorical variable without a reference category (“dummy variable trap”); deriving one variable directly from another (e.g., including height in cm and height in inches).
- ▶ **Imperfect** (but high) collinearity is more common and more pernicious. Coefficients become unstable, standard errors balloon, and signs can flip when a correlated predictor is added or removed. We will diagnose this with VIFs (variance inflation factors) later.

Estimation in MLR

- ▶ The math generalizes: stack outcomes into a vector $\mathbf{y}$ and predictors into a matrix $\mathbf{X}$, then minimize the same quantity as SLR:

$$\text{minimize } \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

- ▶ There is a closed-form OLS solution that generalizes the SLR formulas. R computes it in `lm()`; you do not need to.
- ▶ What *does* matter:
 - ▶ Each $\hat{\beta}_j$ comes with a standard error (in the `summary()` output).
 - ▶ The interpretation of $\hat{\beta}_j$ depends on which other variables are in the model. Different model $\Rightarrow$ different $\hat{\beta}_j$.

Why Diagnostics?

- ▶ Before reporting any inference, check whether the LINE assumptions are plausible.
- ▶ Tool of choice: **residual plots**. A residual is the gap between what we observed and what the model predicted: $e_i = Y_i - \hat{Y}_i$.
- ▶ R produces four standard diagnostic plots automatically: `par(mfrow = c(2,2)); plot(model)`.
- ▶ Principle: check the form of each predictor's relationship with Y *before* fitting (scatterplot with loess smoother), then check residuals *after* fitting.

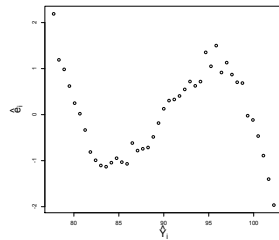
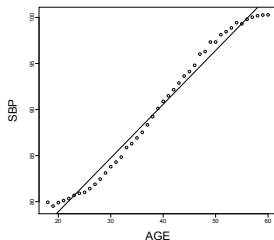
Checking Linearity

- ▶ **Assumption:** the average residual is zero across all combinations of predictors.
- ▶ **How to check:**
 - ▶ Before fitting: plot Y vs. each X with a loess smoother. Does it look roughly straight?
 - ▶ After fitting: plot residuals vs. fitted values. A clear curve flags a linearity problem.

Example violation:

What to do when it fails:

- ▶ Add a polynomial or spline term in X .
- ▶ Transform Y (log, square root) or X .
- ▶ Add an interaction if linearity holds within groups but not overall.



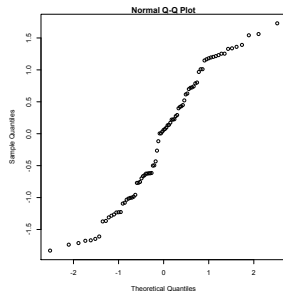
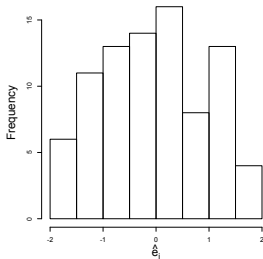
Checking Normality

- ▶ **Assumption:** residuals are normally distributed.
- ▶ **How to check:**
 - ▶ Normal Q-Q plot of residuals. Points should fall close to the diagonal.
 - ▶ Histogram of residuals. Should look roughly bell-shaped.

Example violation:

What to do when it fails:

- ▶ In moderate-to-large samples, mild departures rarely matter.
- ▶ Investigate heavy tails for outliers (data entry errors?).
- ▶ Transform Y (log is most common).



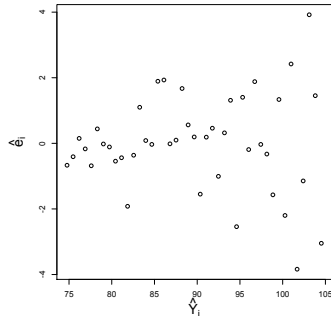
Checking Equal Variance (Homoscedasticity)

- ▶ **Assumption:** residual spread is the same across the range of predictors.
- ▶ **How to check:**
 - ▶ Plot residuals vs. fitted values. Look for a flat band, not a funnel or fan.

Example violation $\implies$

What to do when it fails:

- ▶ Transform Y (log helps when spread grows with the mean).
- ▶ Use weighted least squares.
- ▶ Use robust (sandwich) standard errors, which fix inference without changing $\hat{\beta}$.



The Four Standard Diagnostic Plots in R

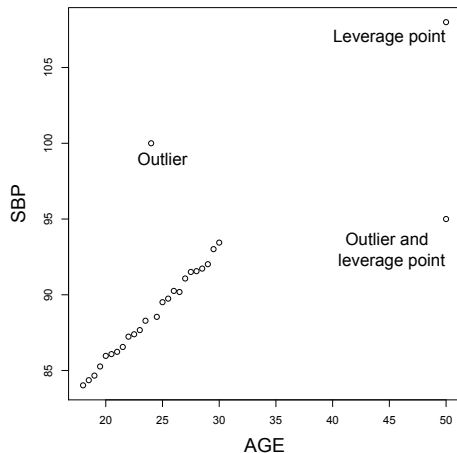
```
par(mfrow = c(2,2)); plot(model)
```

Plot	What to look for
Residuals vs. Fitted	Random scatter around zero. No curve, no funnel.
Normal Q-Q	Points on the diagonal. No heavy tails.
Scale-Location	Flat red line (constant spread).
Residuals vs. Leverage	No points in the upper or lower right corners.

In practice you will spend more time looking at residual plots than any other regression output. Get comfortable reading them.

Outliers, Leverage, and Influence

- ▶ **Leverage point:** unusual predictor values. Far from the center of the X cloud.
- ▶ **Outlier:** unusual outcome value. Large residual.
- ▶ A point can be one, the other, both, or neither.
- ▶ The dangerous combination is high leverage *plus* a large residual. Such a point is *influential*: it visibly pulls the fitted line toward itself, and the inference changes if you remove it.
- ▶ In R : `hatvalues(model)` for leverage, `rstandard(model)` for standardized



Cook's Distance

- ▶ **Cook's distance** quantifies how much the fitted values would shift if a single observation were removed.
- ▶ It blends two things: how unusual the predictors are (leverage) and how badly the model misses on Y (residual size).
- ▶ **How to use it:** plot Cook's distance for every observation and look for points that stand *clearly apart* from the rest. Use the picture, not a formula-based threshold.

Cook's Distance

- ▶ **Cook's distance** quantifies how much the fitted values would shift if a single observation were removed.
- ▶ It blends two things: how unusual the predictors are (leverage) and how badly the model misses on Y (residual size).
- ▶ **How to use it:** plot Cook's distance for every observation and look for points that stand *clearly apart* from the rest. Use the picture, not a formula-based threshold.
- ▶ **Important:** a large Cook's distance flags a point for *investigation*, never automatic removal. Ask:
 - ▶ Is it a data entry or measurement error? Fix it.
 - ▶ Is it a legitimate but extreme observation? Report results with and without it, and let the reader judge.
 - ▶ Is it from a population subgroup your model does not account for? Consider stratifying or adding an interaction.
- ▶ In R : `cooks.distance(model)`; or look at the bottom-right panel of `plot(model)`.

Three Types of Hypothesis Tests in MLR

1. **Global F-test:** $H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0$
 - ▶ “Does the model do anything useful at all?”
 - ▶ Reported at the bottom of `summary(model)`.
2. **Partial t-test for one predictor:** $H_0 : \beta_j = 0$
 - ▶ “Is X_j useful *given the other predictors already in the model?*”
 - ▶ Reported in every row of `summary(model)` output.
3. **Partial F-test for a group of predictors:** $H_0 : \beta_j = \beta_k = \dots = 0$
 - ▶ “Does this group of variables add anything as a set?”
 - ▶ Run with `anova(reduced_model, full_model)`.

Partial t-test: A Conditional Statement

- ▶ The partial t-test for β_j asks a *conditional* question: is X_j informative *on top of* the other variables in the model?
- ▶ Two important consequences:
 - ▶ If two predictors are highly correlated, neither may be significant on its own even when together they explain a lot. Each one looks redundant given the other.
 - ▶ Adding or removing other predictors can flip β_j 's significance, and sometimes its sign.
- ▶ Non-significant does *not* mean “unrelated to Y .” It means “no additional information once the others are in.”
- ▶ This is why a sensible variable-selection strategy considers groups of related predictors, not just individual p-values.

Partial F-test: Comparing Nested Models

- ▶ Used when you want to test a *group* of variables at once, or compare any two nested models (one is a special case of the other).

- ▶ **In R :**

```
anova(reduced_model, full_model)
```

R returns an F-statistic and a p-value. Done.

- ▶ For testing a single variable, the partial F and partial t give exactly the same p-value.
- ▶ **When to prefer the F-test over multiple t-tests:**
 - ▶ Testing all dummy variables for a multi-level categorical predictor jointly.
 - ▶ Testing whether a polynomial or spline expansion of X is needed.
 - ▶ Comparing models that differ by an interaction or a set of related terms.

VO₂Max Example: Partial Tests in Action

Three nested models:

$$(1) \quad VO_2Max = \beta_0 + \beta_1 Age + \epsilon$$

$$(2) \quad VO_2Max = \beta_0 + \beta_1 Runtime + \beta_2 Age + \epsilon$$

$$(3) \quad VO_2Max = \beta_0 + \beta_1 Age + \beta_2 Weight + \epsilon$$

- ▶ **In model 1:** is Age significantly related to VO₂Max? This is the marginal association, ignoring everything else.
- ▶ **In model 2:** is Age significantly related to VO₂Max *after adjusting for Runtime*? Runtime is a strong proxy for fitness, so Age may lose its apparent effect.
- ▶ **In model 3:** is Age significantly related to VO₂Max *after adjusting for Weight*? Weight and Age are less correlated, so Age likely retains its effect.
- ▶ Same predictor (Age), three different conditional questions, potentially three different p-values. None of them is “the right” test on its own. The right test is

Testing a Group of Variables

- ▶ Test whether several predictors are jointly useful, given the others.
- ▶ **VO₂Max example:**

$$\begin{aligned} \text{(Full)} \quad VO_2Max &= \beta_0 + \beta_1 Runtime + \beta_2 Age \\ &\quad + \beta_3 Weight + \beta_4 MaxHR + \epsilon \end{aligned}$$

$$\text{(Reduced)} \quad VO_2Max = \beta_0 + \beta_1 Runtime + \beta_2 Weight + \epsilon$$

- ▶ Question: are Age and MaxHR jointly useful, given Runtime and Weight?
- ▶ In R : `anova(reduced, full)`.
- ▶ A small p-value says “yes, the pair adds something.” A large p-value says “no, you can drop both without losing predictive information.”

Review Questions

- ▶ If we drop a predictor from the model, does R^2 go up, down, or stay the same?
- ▶ If we drop a predictor and R^2 does not change, what does that tell us about that predictor?
- ▶ Can a predictor have a small p-value in one model and a large p-value in a different model that includes more variables? Why?
- ▶ Why is the global F-test sometimes significant when no individual t-test is?
- ▶ **Key takeaway:** R^2 never decreases when you add a predictor, even a useless one. Use adjusted R^2 , AIC, or formal nested-model F-tests when comparing models. We will discuss model selection in more depth in the next session.